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Ex.No:1: Simulation of FDMA in MATLAB

Objective 1:

To simulate an FDMA system by allocating individual frequency channels to multiple users and determine the channel bandwidth assigned to each user.

Program (Matlab Code)

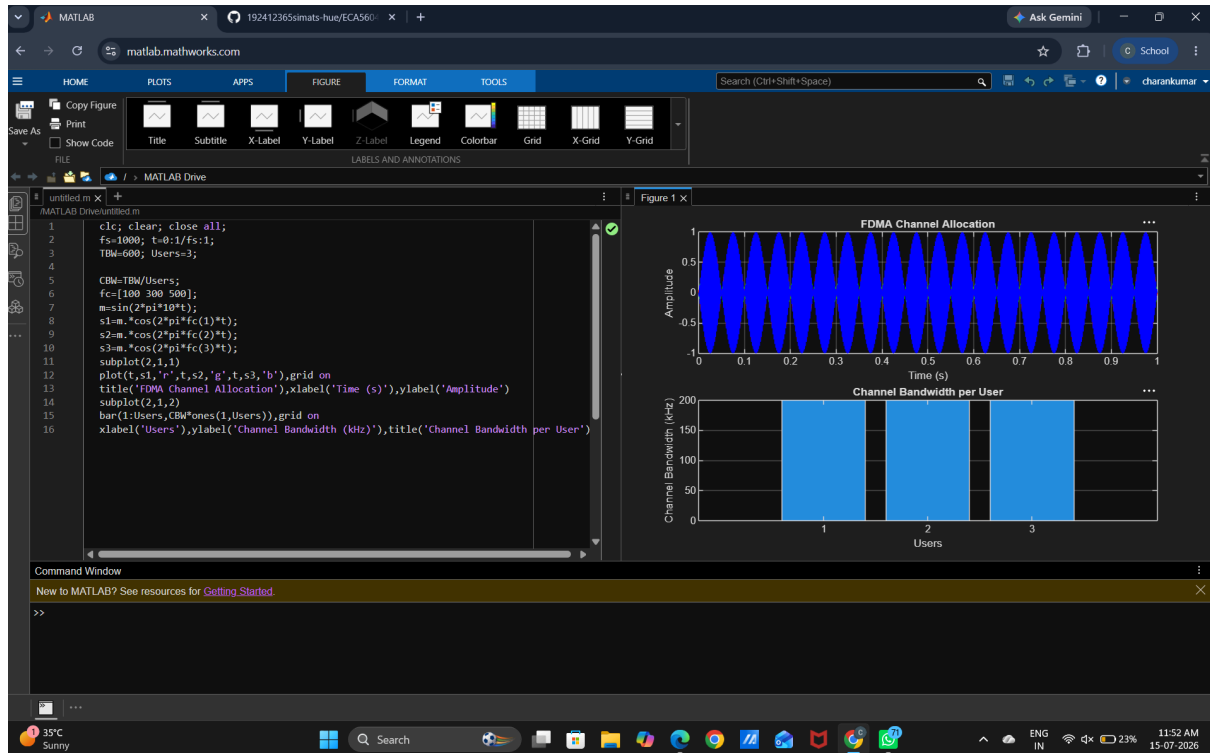
```
clc; clear; close all;
fs=1000; t=0:1/fs:1;
TBW=600; Users=3;

CBW=TBW/Users;
fc=[100 300 500];
m=sin(2*pi*10*t);
s1=m.*cos(2*pi*fc(1)*t);
s2=m.*cos(2*pi*fc(2)*t);
s3=m.*cos(2*pi*fc(3)*t);
subplot(2,1,1)
plot(t,s1,'r',t,s2,'g',t,s3,'b'),grid on
title('FDMA Channel Allocation'),xlabel('Time (s)'),ylabel('Amplitude')
subplot(2,1,2)
bar(1:Users,CBW*ones(1,Users)),grid on
```



xlabel('Users'),ylabel('Channel Bandwidth (kHz)'),title('Channel Bandwidth per User')

Output:



Objective 2:

To calculate the data rate achieved by each user in the FDMA system based on the allocated channel bandwidth.

Matlab Code

```
clc; clear; close all;
```

```
BW = 200; % Channel Bandwidth (kHz)
```

```
Users = 3;
```

```
M = 2; % BPSK Modulation
```

```
DataRate = BW*log2(M); % Data Rate (kbps)
```

```
Rate = DataRate*ones(1,Users);
```

```
disp(' User Data Rate (kbps)')
```

```
disp([(1:Users)' Rate'])
```

```
figure
```

```
bar(1:Users,Rate)
```

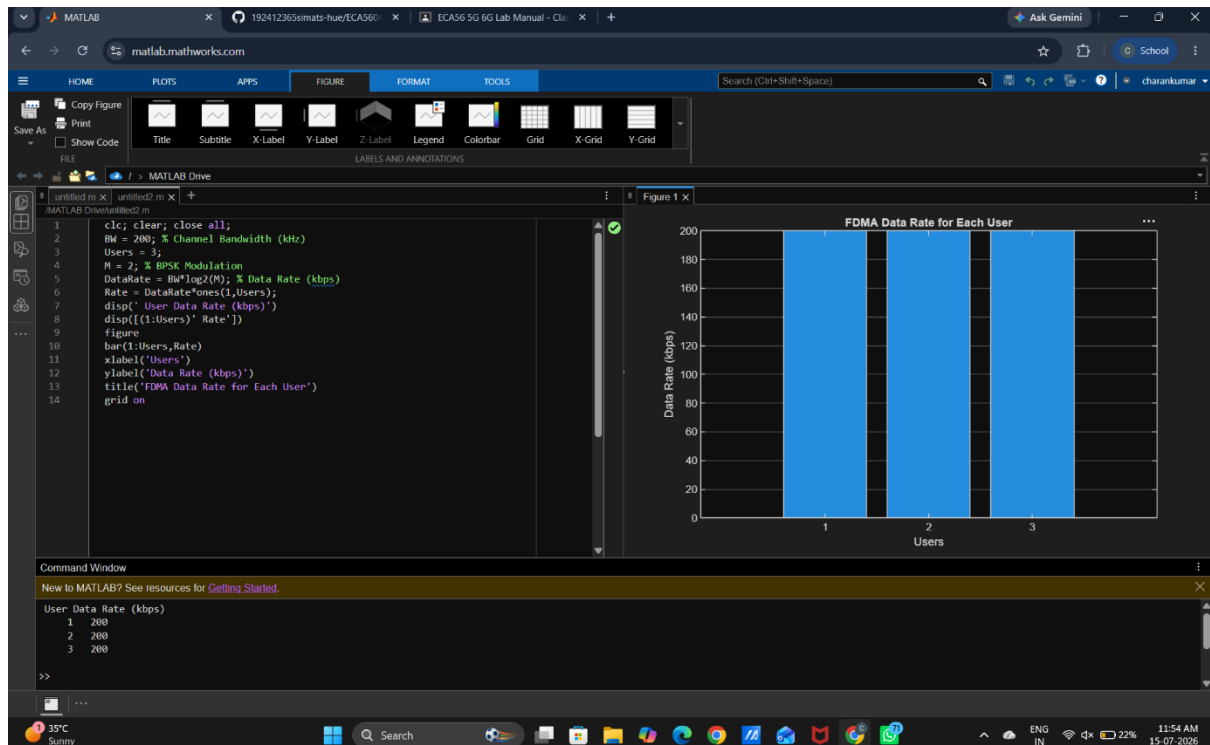
```
xlabel('Users')
```

```
ylabel('Data Rate (kbps)')
```

```
title('FDMA Data Rate for Each User')
```

```
grid on
```

Output



Objective 3:

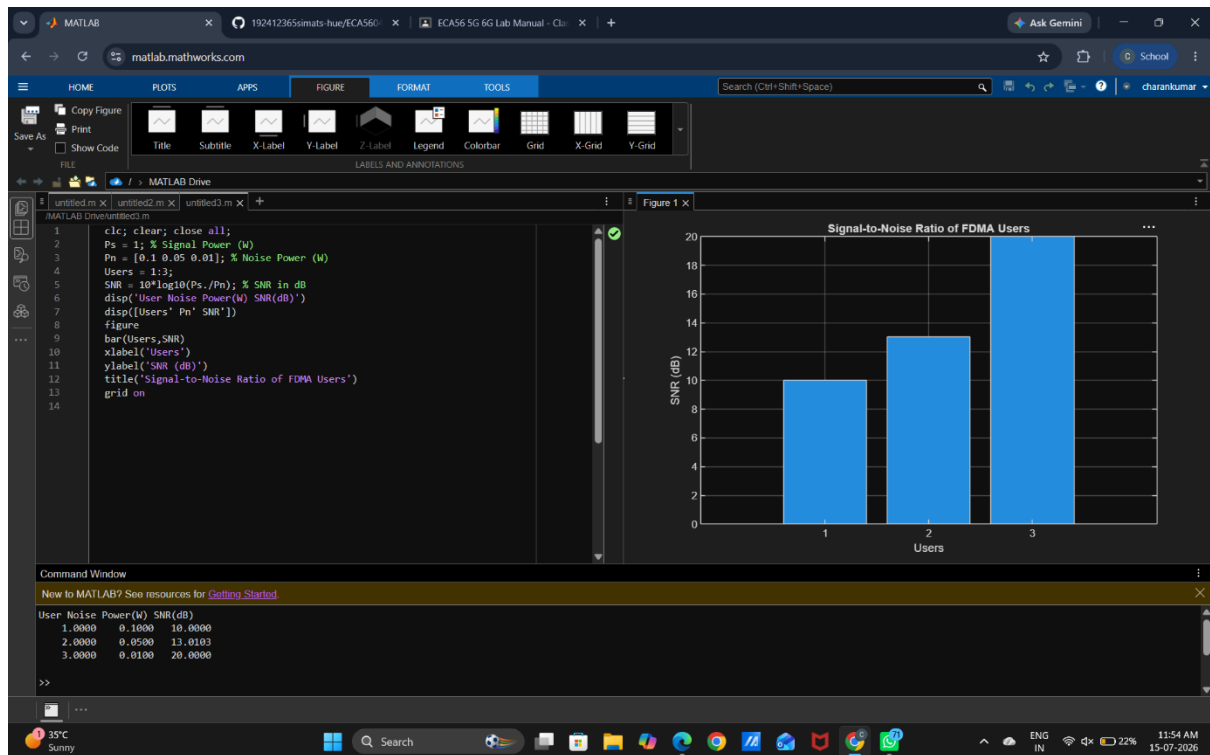
To evaluate the communication quality of the FDMA system by calculating the Signal-to- Noise Ratio (SNR) for different users.

Matlab Code

```
clc; clear; close all;
Ps = 1; % Signal Power (W)
Pn = [0.1 0.05 0.01]; % Noise Power (W)
Users = 1:3;
SNR = 10*log10(Ps./Pn); % SNR in dB
disp('User Noise Power(W) SNR(dB)')
disp([Users' Pn' SNR'])
figure
bar(Users,SNR)
xlabel('Users')
ylabel('SNR (dB)')
title('Signal-to-Noise Ratio of FDMA Users')
grid on
```



Output



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